

DRAFT: The Scalar-Angular-Torsion Framework: VIII. Comprehensive Glossary of the Zottenwelt

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We present a formalized glossary of terms and constants for the Scalar-Angular-Torsion (SAT) framework. This document defines the ontological, topological, and mechanical primitives required to maintain the Zero-Parameter Economy. By anchoring terms to the 24-cell HSUCV lattice geometry, we establish the metrological basis for Failure Rigidity and the Universal Indicatrix (UI) mapping engine.

LAY OVERVIEW

The SAT framework uses many specific terms that have unique, mechanical meanings derived from four-dimensional geometry. In this model, **Mass** is not a substance but a type of “drag” called **Projective Resistance**. **Gravity** is not a separate force but a “faint hum” or **Collective Attenuation** of much stronger forces happening in four dimensions.

This glossary acts as the rulebook for the **Zero-Parameter Economy**. It defines the rigid “gears” of the universe, such as the **Projection Constant** (B), which sets the scale for everything from the weight of an atom to the speed of light. By defining these terms precisely, the framework ensures **Failure Rigidity**: every definition is so tightly linked to the 4D geometry that the theory cannot be adjusted without breaking the entire mathematical system.

I. ONTOLOGICAL AND SUBSTRATE TERMS

24-Cell HSUCV Lattice: The Hypersphere Unit Cell Vertex lattice; the discretized 4D coordinate grid serving as the universal frame of reference for the expanding hypersphere.

Failure Rigidity: The principle that the framework is structurally locked; any modification to base anchors ripples catastrophically through all derived sectors, ensuring the theory is either entirely correct or fundamentally broken.

Filamental Primacy: The ontological postulate that 4D worldline filaments are the primary physical units of the universe, possessing intrinsic tension T .

Intersectional Trace: Perceived 3D particles are the statistical intersections where 4D filaments pierce the propagating 3-sphere “time surface” Σ_t .

Zero-Parameter Economy: A theoretical regime where all physical constants emerge as mandatory geometric invariants of the 4D substrate, eliminating manual tuning.

Zottenwelt: The foundational substrate of physics; a hypercomplex tangled fabric of physical 4D worldline filaments.

II. FUNDAMENTAL CONSTANTS AND ANCHORS

Filament Scale (l_f): The universal measuring rod (≈ 0.7937 fm) derived from the topological saturation limit; pinned to the Rydberg Constant.

Obscuration Constant (θ_{obs}): The “Geometric Corner” or blackout threshold (≈ 0.246 rad or 14.1°) where 3D gauge coupling (visibility) vanishes.

Projection Constant (B): The fundamental geometric fingerprint (≈ 0.2387 rad) representing the per-dimension share of 4D-to-3D projection distortion ($3/4\pi$).

Speed of Time (c): The radial expansion rate of the 4D hypersphere (S^3), manifesting in 3D as the speed of light.

Universal Mass Anchor (m_0): The bare filament mass scale ($\approx 1.0073 \times 10^{-27}$ kg) derived from intrinsic filament tension.

III. TOPOLOGICAL AND MECHANICAL TERMS

Achromatic Phase Snap: A discrete, frequency-independent jump (≈ 0.246 rad) in timing residuals caused by a worldline “clicking” to the next lattice node at v_{crit} .

Braid-Smoothing Metric: A non-linear geometric correction for high-density inter-braiding; resolves the ≈ 26 – 27% mass deviation in high-density nuclei like Helium-3.

Collective Attenuation: The mechanical origin of gravity; the process where raw 4D filament tension ($G_{raw} \approx 1.2 \times 10^{44}$ N) is filtered through the lattice topological mode density ($\rho_{embed} \approx 10^{-19}$).

Critical Velocity (v_{crit}): The velocity threshold ($B \cdot c \approx 0.2387c$) where matter begins to precess toward the Obscuration Constant.

Dark Matter Illusion: A geometric effect where matter precessed past θ_{obs} remains gravitationally active but loses observable 3D gauge coupling.

Geometric Blackout: The state of a filament that has rotated past the Obscuration Constant, becoming invisible in the 3D slice.

Misalignment Angle (θ_4): The angle between a filament tangent and the radial time-flow vector; the geometric origin of relativistic phenomena.

Projective Resistance (R): The mechanical drag a 4D filament encounters against the expansion of the universe; the SAT reinterpretation of mass.

Topological Charge (Q): The integrated knot complexity (linking and winding numbers) of a particle's 4D worldline.

Universal Indicatrix (UI): The mapping engine translating 4D hyperhelical filamental paths into observable 3D trajectories.

UV Finiteness Lock: The structural constraint limiting stable ground-state intersections to a maximum of three filaments per lattice node ($Q \leq 3$).

IV. PARTICLE TAXONOMY

True Bosons: Transient, non-persistent filamental excitations (e.g., photons, neutrinos) lacking a persistent superhelical identity.

True Fermions: Particles with persistent 4D coiling histories (e.g., leptons, mesons, baryons) that maintain identity across the lattice.